

Indices and surds - Set 1 - Solutions

Specification MM1.1, MM1.2, M2.7, M2.11, M2.12.

Answer key

Question	Answer	Topic
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9	A	Indices and surds
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Worked solutions

Question 1 IND-01 · Indices and surds · spec MM1.1

Simplify $\frac{(2x^3)^4 \times 8x^{-5}}{4x^2}$.

Answer: F

$(2x^3)^4 = 16x^{12}$, so the numerator is $16x^{12} \times 8x^{-5} = 128x^7$. Dividing by $4x^2$ gives $32x^5$.

Full method - how and why

A pure index-laws question. Nothing here needs thought once you treat the numbers and the powers of x as two separate bookkeeping jobs.

1 Deal with the bracket first: $(2x^3)^4 = 2^4 x^{12} = 16x^{12}$.

why: A power outside a bracket multiplies **every** factor inside, so the 2 is raised to the 4th power as well as the x^3 . This is the step people skip.

2 Multiply the numerator: $16x^{12} \times 8x^{-5} = 128x^7$.

why: Multiplying powers of the same base adds the indices: $12 + (-5) = 7$. The numbers just multiply normally.

3 Divide by $4x^2$: $\frac{128x^7}{4x^2} = 32x^5$.

why: Dividing subtracts indices: $7 - 2 = 5$, and $128 \div 4 = 32$.

Where it goes wrong

- Forgetting to raise the 2 to the 4th power gives $2x^{12}$ at step 1 and the answer $4x^5$.
- Subtracting instead of adding the -5 gives x^9 and the answer $32x^9$ - which is offered.

Takeaway Handle the number part and the index part separately, and always distribute an outside power over everything in the bracket.

Question 2 IND-02 · Indices and surds · spec MM1.1

Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$.

Answer: C

The negative index inverts the fraction: $\left(\frac{8}{27}\right)^{2/3}$. Cube root first: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

Full method - how and why

A fractional **negative** index of a fraction. Three separate instructions are packed into $-\frac{2}{3}$: invert, take the cube root, square.

1 The minus sign inverts the fraction: $\left(\frac{27}{8}\right)^{-2/3} = \left(\frac{8}{27}\right)^{2/3}$.

why: $a^{-n} = \frac{1}{a^n}$, and the reciprocal of a fraction is the fraction turned upside down. Doing this first keeps the numbers small.

2 Take the cube root of top and bottom: $\left(\frac{8}{27}\right)^{1/3} = \frac{2}{3}$.

why: $a^{m/n} = (\sqrt[n]{a})^m$; rooting before powering keeps you working with 2 and 3 rather than 64 and 729.

3 Square the result: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

why: The numerator 2 in the index is the power.

Where it goes wrong

- Treating the minus sign as making the answer negative gives $-\frac{9}{4}$ or $-\frac{4}{9}$; a negative index never makes a positive number negative.
- Forgetting to square gives $\frac{2}{3}$; forgetting to invert gives $\frac{9}{4}$. All three are offered as options.

Takeaway Read a fractional index as three instructions - sign (invert), denominator (root), numerator (power) - and do them in that order.

Question 3 IND-03 · Indices and surds · spec MM1.1

Solve $4^{x+1} = 8^{2x-3}$.

Answer: A

Write both sides as powers of 2: $2^{2x+2} = 2^{6x-9}$. So $2x+2 = 6x-9$, giving $4x = 11$ and $x = \frac{11}{4}$.

Full method - how and why

An exponential equation where both sides can be written as powers of the **same** base. Once the bases match, the indices must be equal.

- 1 Spot that $4 = 2^2$ and $8 = 2^3$.

why: Both numbers are powers of 2, so the whole equation can be rewritten in base 2 and the exponentials compared directly.

- 2 Rewrite: $4^{x+1} = 2^{2(x+1)} = 2^{2x+2}$ and $8^{2x-3} = 2^{3(2x-3)} = 2^{6x-9}$.

why: $(a^m)^n = a^{mn}$; the outer index multiplies the whole inner index, so brackets are essential.

- 3 Equate the indices: $2x+2 = 6x-9$, so $4x = 11$ and $x = \frac{11}{4}$.

why: $2^u = 2^v$ forces $u = v$ because 2^x is a one-to-one function (it never takes the same value twice).

Where it goes wrong

- Multiplying out as $3 \times 2x - 3$ instead of $3(2x - 3)$ loses the bracket and gives a different linear equation.
- Writing $8 = 2^4$ (a slip) gives $x = \frac{7}{2}$, which is offered.

Takeaway Whenever an equation has different bases, look for a common base first; comparing indices is then legitimate because exponentials are one-to-one.

Question 4 IND-04 · Indices and surds · spec MM1.2

Express $\frac{5}{3+2\sqrt{5}}$ in the form $a + b\sqrt{5}$ where a and b are rational.

Answer: B

Multiply top and bottom by $3 - 2\sqrt{5}$: denominator $9 - 20 = -11$, numerator $15 - 10\sqrt{5}$. So the value is $\frac{15 - 10\sqrt{5}}{-11} = \frac{-15 + 10\sqrt{5}}{11}$.

Full method - how and why

Rationalising a denominator of the form $p + q\sqrt{n}$. The tool is the difference of two squares, which turns the surd denominator into an integer.

- 1 Multiply numerator and denominator by the conjugate $3 - 2\sqrt{5}$.

why: $(p + q\sqrt{n})(p - q\sqrt{n}) = p^2 - q^2n$, which contains no surd. Multiplying top and bottom by the same thing does not change the value.

- 2 Denominator: $3^2 - (2\sqrt{5})^2 = 9 - 4 \times 5 = -11$.

why: Square the **whole** second term, coefficient included: $(2\sqrt{5})^2 = 4 \times 5$, not 2×5 .

- 3 Numerator: $5(3 - 2\sqrt{5}) = 15 - 10\sqrt{5}$.

why: Nothing subtle - just expand.

- 4 Tidy the signs: $\frac{15 - 10\sqrt{5}}{-11} = \frac{-15 + 10\sqrt{5}}{11}$.

why: A negative denominator is not 'in the form $a + b\sqrt{5}$ ' with the usual convention, and both versions are offered, so the sign handling decides the mark.

Where it goes wrong

- $(2\sqrt{5})^2 = 10$ (forgetting to square the 2) gives denominator -1 and a wildly different answer.
- Leaving $\frac{15 - 10\sqrt{5}}{-11}$ and picking the option with $+11$ in the denominator but the wrong signs on top.

Takeaway Multiply by the conjugate, square the coefficient as well as the surd, then fix the sign of the denominator at the end.

Question 5 IND-05 · Indices and surds · spec MM1.2

Simplify $\sqrt{75} - \sqrt{27} + \sqrt{12}$.

Answer: E

$$\sqrt{75} = 5\sqrt{3}, \sqrt{27} = 3\sqrt{3}, \sqrt{12} = 2\sqrt{3}, \text{ so the total is } (5 - 3 + 2)\sqrt{3} = 4\sqrt{3}.$$

Full method - how and why

Simplifying a sum of surds. They can only be combined once each is written as a multiple of the **same** surd.

- 1 Pull out the largest square factor from each: $\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$, $\sqrt{27} = \sqrt{9 \times 3} = 3\sqrt{3}$,
 $\sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3}$.
why: $\sqrt{ab} = \sqrt{a}\sqrt{b}$, so a perfect-square factor comes out as a whole number. All three reduce to multiples of $\sqrt{3}$, which is why the sum simplifies at all.
- 2 Collect like terms: $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3} = 4\sqrt{3}$.
why: Once every term has the same surd part, $\sqrt{3}$ behaves exactly like an algebraic x .

Where it goes wrong

- Adding under one root sign: $\sqrt{75} - \sqrt{27} \neq \sqrt{48}$. Roots do not distribute over $+$ or $-$.
- Using $\sqrt{75} = 3\sqrt{5}$ (wrong square factor) - always take out the **largest** square.

Takeaway Surds add only when their irrational parts match, so simplify each one fully before trying to combine.

Question 6 IND-08 · Indices and surds · spec MM1.1

Solve $9^x - 4 \times 3^x + 3 = 0$.

Answer: B

$$\text{Let } y = 3^x: y^2 - 4y + 3 = (y - 1)(y - 3) = 0, \text{ so } 3^x = 1 \text{ or } 3^x = 3, \text{ giving } x = 0 \text{ or } x = 1.$$

Full method - how and why

A hidden quadratic. The key observation is $9^x = (3^2)^x = (3^x)^2$, so the equation is quadratic in 3^x .

- 1 Write $9^x = (3^x)^2$ and substitute $y = 3^x$: $y^2 - 4y + 3 = 0$.
why: Spotting that one exponential is the square of the other is the whole question; without it there is no route in.
- 2 Factorise: $(y - 1)(y - 3) = 0$, so $y = 1$ or $y = 3$.
why: Ordinary quadratic technique once the substitution is made.
- 3 Undo the substitution: $3^x = 1 \Rightarrow x = 0$; $3^x = 3 \Rightarrow x = 1$.
why: $a^0 = 1$ for any a - the root $y = 1$ is easy to lose because it looks trivial.

Where it goes wrong

- Stopping at $y = 1, 3$ and offering those as the answers - the question asks for x , not 3^x .
- Rejecting $y = 1$ as 'not a real solution'. Only **non-positive** values of y must be rejected, because $3^x > 0$ always.

Takeaway If an equation contains a^{2x} and a^x , substitute $y = a^x$; afterwards discard only the roots with $y \leq 0$.

Question 7 IND-10 · Indices and surds · spec MM1.1

The value of $\frac{2^{n+2} + 2^n}{2^{n+1}}$ is

Answer: E

Factor 2^n : numerator $= 2^n(4 + 1) = 5 \cdot 2^n$; denominator $= 2 \cdot 2^n$. Quotient $= \frac{5}{2}$.

Full method - how and why

Powers with an unknown exponent. Factorising out the smallest power turns the expression into a number.

- 1 Write every term with the common factor 2^n : $2^{n+2} = 4 \cdot 2^n$, $2^{n+1} = 2 \cdot 2^n$.

why: $2^{n+k} = 2^k \cdot 2^n$. Choosing the *smallest* power (2^n) as the common factor makes every remaining factor a whole number.

- 2 Substitute: $\frac{4 \cdot 2^n + 2^n}{2 \cdot 2^n} = \frac{5 \cdot 2^n}{2 \cdot 2^n}$.

why: Factorising the numerator is legitimate because both terms contain 2^n .

- 3 Cancel 2^n : the value is $\frac{5}{2}$, independent of n .

why: $2^n \neq 0$ for every n , so cancelling is always valid here.

Where it goes wrong

- 'Cancelling' the n 's in the indices, e.g. claiming $\frac{2^{n+2}}{2^{n+1}} = 2^2/2^1$ term by term across a sum - you cannot split a fraction over a sum in the denominator.
- Answering 2^n because the n 'has to appear somewhere'.

Takeaway With unknown exponents, factor out the lowest power; the n almost always cancels and the answer is a plain number.

Question 8 IND-13 · Indices and surds · spec MM1.1

Simplify $\left(\frac{a^{\frac{1}{2}}b^{-2}}{a^{-1}b^{\frac{1}{2}}}\right)^4$.

Answer: E

Inside the bracket: $a^{\frac{1}{2}+1}b^{-2-\frac{1}{2}} = a^{\frac{3}{2}}b^{-\frac{5}{2}}$. Raise to the 4th power: a^6b^{-10} .

Full method - how and why

Index laws with two variables. Simplify inside the bracket first, then apply the outside power to each index.

- 1 Simplify the a 's inside: $\frac{a^{\frac{1}{2}}}{a^{-1}} = a^{\frac{1}{2}-(-1)} = a^{\frac{3}{2}}$.

why: Dividing subtracts indices, and subtracting a negative adds - the commonest slip in the question.

- 2 Simplify the b 's: $\frac{b^{-2}}{b^{\frac{1}{2}}} = b^{-2-\frac{1}{2}} = b^{-\frac{5}{2}}$.

why: Same rule; keep the two variables in separate columns so signs do not get mixed up.

- 3 Raise the bracket to the 4th power: $(a^{\frac{3}{2}}b^{-\frac{5}{2}})^4 = a^6b^{-10}$.

why: $(a^m)^n = a^{mn}$ applies to each base separately: $\frac{3}{2} \times 4 = 6$ and $-\frac{5}{2} \times 4 = -10$.

Where it goes wrong

- Getting $a^{-1/2}$ at step 1 by subtracting the wrong way round, leading to $a^{-2}b^{10}$ - which is offered.
- Applying the power 4 to only the first factor.

Takeaway Simplify inside the bracket to a single power of each base before applying the outer index.

Question 9 IND-14 · Indices and surds · spec MM1.2

The number $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$ is equal to

Answer: A

$$\sqrt{50} = 5\sqrt{2}, \sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}. \text{ So } \frac{8\sqrt{2}}{2\sqrt{2}} = 4.$$

Full method - how and why

A quotient of surds that collapses completely. Simplify each root first and the $\sqrt{2}$'s cancel.

1 Simplify: $\sqrt{50} = 5\sqrt{2}, \sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}$.

why: Every number is (square) $\times 2$, so every root is a multiple of $\sqrt{2}$ - which is why the fraction will be rational.

2 Add the numerator: $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$.

why: Like terms again - $\sqrt{2}$ behaves as a common factor.

3 Divide: $\frac{8\sqrt{2}}{2\sqrt{2}} = 4$.

why: The $\sqrt{2}$ cancels top and bottom, leaving plain arithmetic.

Where it goes wrong

- Working out $\sqrt{50} + \sqrt{18} = \sqrt{68}$, which is wrong and leads nowhere.
- Cancelling only the numbers and leaving a stray $\sqrt{2}$, giving $4\sqrt{2}$.

Takeaway Simplify every surd to the form $k\sqrt{m}$ before adding or dividing; common surd factors then cancel visibly.

Question 10 IND-15 · Indices and surds · spec MM1.1

Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .

Answer: C

$$2^{2x-y} = \frac{(2^x)^2}{2^y} = \frac{25}{20} = \frac{5}{4}.$$

Full method - how and why

You are given two exponentials and asked for a third. Express the target in terms of the given ones using index laws - you never need to find x or y .

1 Split the index: $2^{2x-y} = \frac{2^{2x}}{2^y}$.

why: Subtraction in the index is division - this is what lets you use the two given facts separately.

2 Write $2^{2x} = (2^x)^2 = 5^2 = 25$.

why: A coefficient in the index is a power of the whole exponential: $2^{2x} = (2^x)^2$, not 2×2^x .

3 Substitute: $\frac{25}{20} = \frac{5}{4}$.

why: $2^y = 20$ is given directly.

Where it goes wrong

- Using $2^{2x} = 2 \times 2^x = 10$, giving $\frac{1}{2}$.
- Trying to solve for $x = \log_2 5$ and $y = \log_2 20$ first - correct but slow, and it invites arithmetic slips.

Takeaway Rewrite the target index as sums and multiples of the given ones; substitution then beats solving for the unknowns.

Question 11 IND-17 · Indices and surds · spec MM1.1

Solve $x^{\frac{3}{2}} = 8x^{-\frac{3}{2}}$ for $x > 0$.

Answer: E

Multiply by $x^{3/2}$: $x^3 = 8$, so $x = 2$.

Full method - how and why

An equation with the unknown in two fractional powers. Multiplying by the negative power clears it entirely.

1 Multiply both sides by $x^{3/2}$: $x^{3/2} \cdot x^{3/2} = 8$, i.e. $x^3 = 8$.

why: $x^{-3/2}$ becomes $x^0 = 1$ on the right, and the left gains $\frac{3}{2} + \frac{3}{2} = 3$. Multiplying by a positive quantity (we are told $x > 0$) cannot introduce false solutions.

2 Take the cube root: $x = 2$.

why: $x^3 = 8$ has one real root.

Where it goes wrong

- Cancelling the powers to get $x = 8$.
- Dividing by $x^{3/2}$ instead of multiplying, which leaves $x^3 = 8$ on the wrong side and invites sign errors in the index.

Takeaway To clear a negative fractional power, multiply through by that power - stated positivity guarantees the step is reversible.

Question 12 IND-19 · Indices and surds · spec MM1.1

How many real solutions does the equation $2^{x^2} = 16 \times 2^{-3x}$ have?

Answer: B

$2^{x^2} = 2^{4-3x}$, so $x^2 + 3x - 4 = 0$, i.e. $(x + 4)(x - 1) = 0$: two real solutions.

Full method - how and why

'How many real solutions' with exponentials - convert to a single base and the exponents give a quadratic whose discriminant answers the question.

1 Write $16 = 2^4$, so the right-hand side is $2^4 \cdot 2^{-3x} = 2^{4-3x}$.

why: Multiplying powers of the same base adds indices; this puts both sides in base 2.

2 Equate indices: $x^2 = 4 - 3x$, i.e. $x^2 + 3x - 4 = 0$.

why: Legitimate because 2^u is one-to-one.

3 Factorise: $(x + 4)(x - 1) = 0$ - two distinct real roots.

why: The question asks only for the *number* of solutions, so the discriminant ($9 + 16 > 0$) would do; factorising confirms both roots are real.

Where it goes wrong

- Rejecting $x = -4$ because it is negative - the **exponent** may be negative; it is 2^x that must stay positive.
- Reading 16×2^{-3x} as $(16 \times 2)^{-3x}$.

Takeaway Same base $\Rightarrow$ equate exponents; then it is an ordinary quadratic and every real root counts.